

Phantom Walls : Spatial perception and navigation without vision

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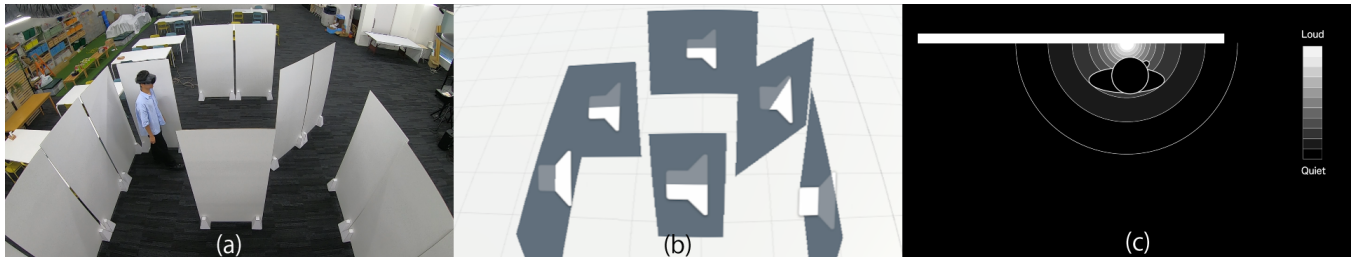


Figure 1: Phantom Walls is a new method for establishing continuous spatial perception and navigation without vision. (a) Walking while avoiding walls. For illustrative purpose physical walls are aligned, that are not visible to the user. (b) Phantom walls placed in VR space (c) Three laws that govern this auditory world filled with phantom walls

ABSTRACT

"Phantom Walls" is a novel technique that establishes continuous spatial perception without vision. By creating an auditory environment where obstacles emit sounds, users can perceive and navigate around visually imperceptible "Phantom" obstacles by listening to the generated "soundscapes". This method allows individuals to perceive and avoid these obstacles while walking without vision.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*.

KEYWORDS

spatial perception, sensory substitution, virtual reality, navigation

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1 INTRODUCTION

In the past, various methods of non-visual obstacle perception have been proposed including echolocation, which utilizes environmental sounds and the echoes of self-generated sounds to perceive the

spatial configuration of obstacles[Kolarik et al. 2014], as well as contact methods involving tools like white canes. Sensory substitution techniques, which utilizes senses such as sound and touch, have also been used to replace visual information[Chebat et al. 2015; Neugebauer et al. 2020].

In this study, we propose a novel approach called "Phantom Walls" which is a blind navigation technique that enables the perception of spatial information at all times in time and space by using VR technology. By applying cross-modal effects from spatial sound to somatic feeling, phantom pressure is generated from invisible walls to achieve a smooth and intuitive spatial perception. Phantom Walls consist of invisible "phantom" walls placed in a virtual space. Paired with virtual speakers, users can listen to the sounds of the walls. By perceiving the auditory world simulated within this space, users can navigate while avoiding the unseen phantom walls.

2 DESIGN AND IMPLEMENTATION

Within the virtual space that constructs the auditory world, virtual speakers are positioned to represent the continuous relationship between the user and the phantom walls through their movement, changes in volume, and playback speed based on the user's movement. The following principles are specifically applied to create the auditory world, with a visual representation provided in Figure1(c).

- (1) Each obstacle is associated with a single speaker, and the speaker moves to the location on the obstacle's surface that is closest to the user. This property allows for the perception of a wall on the right side. For example, if the user consistently hears sound from their right side while walking straight, they can better identify this object.
- (2) The volume of the sound increases as the distance to the obstacle decreases. When the user gets closer to an obstacle, the sound appears louder, creating a sensation of "pressure" or "rebound". The relationship between distance and sound

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is adjusted as shown in Figure2 to enhance the navigation experience.

- (3) The playback speed of the sound varies based on the angle between the user's head orientation and the wall. When the user's head is perpendicular to the wall, the sound is played back at a slower speed, while it is played back at a faster speed when the head orientation is parallel to the wall. The playback speed of the sound changes linearly along with the angle.

By applying these principles, users can perceive the auditory world and navigate their surrounding space by listening to the "sound field" while walking and rotating their head.

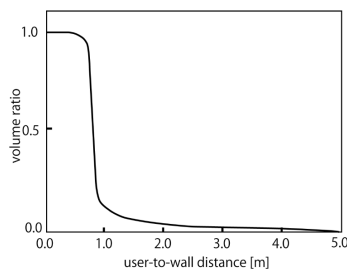


Figure 2: The relationship between distance and volume tuned to walk while avoiding static walls

3 APPLICATION AND FUTURE WORK

We implemented phantom walls, which is not necessarily match the physical objects in physical environments. By matching them, however, this method can be applied to a new method for the navigation in physical environments. To apply to physical environments to unknown spatial configurations and moving objects, we need to integrate real-time mapping and tracking technologies into the system (e.g., SLAM: Simultaneous Localization and Mapping) (Figure3). If this method can be applied to real-world navigation without vision, then we will explore more in detail about the application for the navigation method for people with visual impairment (Figure4). This is a definitely significant next step of our research.

Furthermore, for non-blind people, the system can be applied as room-scale walking VR entertainment without using vision, and that will be a demonstration in the SIGGRAPH Asia 2023, which is described below.

4 USER EXPERIENCE

At SIGGRAPH Asia 2023, participants wearing a blindfold and a head-mounted display (Meta Quest Pro) will have the opportunity to experience perceiving and avoiding "phantom" walls that are not visible to the naked eye by listening to the "sound field" in a demo space. Additionally, there are plans to incorporate the experience of navigating around invisible moving obstacles placed within the virtual world.

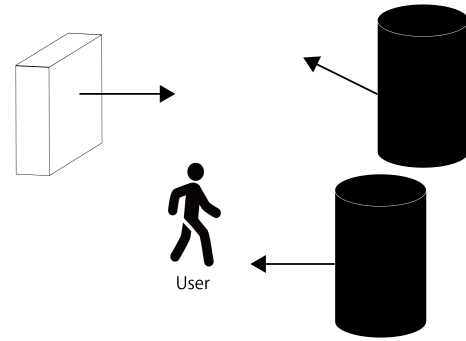


Figure 3: Navigation in an environment with moving objects

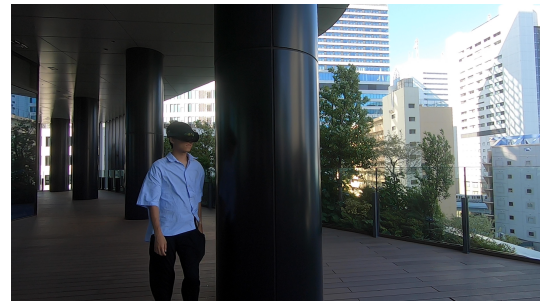


Figure 4: By combining real-time mapping technologies with the system, it becomes possible to apply the concept to physical environments as well.

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